Evidence-Calibrated Enterprise RAG for Hallucination-Resistant Question Answering

Abayomi Adelowotan
Chevron
Lagos, Nigeria
abayomi.adelowotan@chevron.com

Kikelomo Obayemi
Chevron
Lagos, Nigeria
liim@chevron.com

***Abstract*—Retrieval-augmented generation (RAG) reduces but does not remove unsupported answers in enterprise question answering, where evidence may be incomplete, contradictory, or outdated. We study Evidence-Calibrated Enterprise RAG (ECERAG), a selective answering framework that scores the sufficiency of retrieved evidence from four signals—relevance, coverage, agreement, and source trust/recency—and answers, asks for clarification, or abstains accordingly. We evaluate it on a 60-question benchmark built from SEC filings, the NIST AI Risk Management Framework, the OWASP Top 10, and current/obsolete IETF specifications, under clean retrieval and three perturbations (irrelevant, contradictory, and outdated evidence), with five open-weight generators from 0.5B to 14B parameters, repeated cross-validation, paired bootstrap intervals, and two independent automatic graders checked against a human annotator. Relative to a RAG baseline that must always answer, ECERAG cuts the error rate on answered questions by roughly two-thirds (e.g., 0.43 to 0.13 at 7B; 0.40 to 0.11 at 14B). A controlled decomposition shows that this gain comes almost entirely from the generator's own evidence-grounded refusal: ECERAG is not significantly better than self-refusal alone under the primary grader, and the external sufficiency gate on its own yields small reductions (at most 0.01–0.11), significant only for the 7B model or under outdated evidence. The sufficiency score is nevertheless a reliable ranking signal: among answered questions it orders wrong answers below right ones better than chance at every generator size from 1.5B to 14B. We conclude that *retrieve–assess–decide* is effective, that assessment is best led by grounded self-refusal, and that evidence-side scores are a complementary confidence signal rather than a standalone defense.**

***Index Terms*—retrieval-augmented generation, hallucination, selective prediction, calibration, enterprise question answering, trustworthy AI**

I. Introduction

Enterprise deployments of large language models (LLMs) need factual reliability and auditable behavior under uncertainty: in financial, legal, and engineering settings a confident wrong answer can cause material harm. Retrieval-augmented generation (RAG) grounds answers in retrieved documents [1], [2], but a standard pipeline still answers unconditionally, even when the retrieved passages are irrelevant, contradictory, or superseded.

ECERAG makes answering conditional on an explicit assessment of the evidence. A sufficiency score $S(q, E)$ routes each query to *Answer*, *Clarify*, or *Abstain*, in the tradition of selective prediction [3], [4]. Our research question is: *can evidence-sufficiency signals reduce unsupported answers in enterprise RAG without materially degrading coverage, and how much of any reduction is due to the evidence gate itself?*

A pilot of this design with 40 questions and a 0.5B-parameter generator found no reduction in selective risk, and attributed most errors to the generator. This paper replaces that pilot with a controlled study and makes five contributions:

1) A fully specified sufficiency score and selective policy (Sec. III), including a conflict check that detects near-duplicate passages disagreeing on a figure.
2) A generator-scale study (0.5B–14B) with repeated cross-validation, so every question is a held-out test item, and paired bootstrap confidence intervals (Sec. IV).
3) A decomposition that separates the evidence gate from the generator's own refusal by adding a truly unconditional baseline and a gate-only system (Sec. V-B).
4) Evidence that the sufficiency score is a reliable *ranking* signal for answer correctness across scales, even where thresholded gating adds little (Sec. V-D).
5) A human check of both automatic graders on 100 answers, which supports the LLM judge as the primary grader and bounds how small an effect the automatic grading can resolve (Sec. V-G).

II. Related Work

Retrieval-augmented generation. RAG couples a parametric generator with non-parametric retrieval [1], building on dense retrieval [5], late interaction [6], and fusion-in-decoder readers [7]; surveys document persistent failures under weak or noisy retrieval [2]. **Adaptive and corrective retrieval.** Self-RAG [8] trains the generator to retrieve and critique on demand; Corrective RAG [9] detects poor retrieval and re-retrieves. Neither defines an explicit abstention policy calibrated to enterprise risk. Closest to our work, Joren et al. [10] classify whether retrieved context is sufficient with an LLM rater and combine that label with model confidence to decide when to abstain. ECERAG instead builds sufficiency from cheap, inspectable retrieval-side signals (including document version and conflicts between passages), and our decomposition measures how much such a gate adds over the generator's own refusal. **Selective prediction.** A selective model trades coverage for lower risk on the inputs it accepts [3], [11]; Kamath et al. [4] apply this to QA under domain shift, and LLMs have been shown to partly know what they know [12]—a point our decomposition

makes concrete for RAG. Generator-side uncertainty such as semantic entropy [13] detects confabulation without looking at the evidence; we treat it as complementary and leave a direct comparison to future work. **Evaluation.** RAG evaluation has moved toward faithfulness and context relevance [14], robustness to retrieval shift [15], and attribution [16]; LLM-as-judge grading [17] is increasingly used and should be cross-checked, as we do here.

III. Method

A. Retrieval

Documents are split into 220-word passages with a 40-word overlap. For a query q, BM25 [18] scores and cosine similarities of `all-MiniLM-L6-v2` embeddings [19] are each min–max normalized over the corpus and averaged with equal weight. The top 20 passages are reranked with the `ms-marco-MiniLM-L-6-v2` cross-encoder [20], [21], and the top five form the evidence set $E = \{e_1, \ldots, e_5\}$.

B. Evidence Sufficiency Score

$S(q, E) = \frac{1}{4}(R + C + A + T) \in [0, 1]$, with equal weights (fitting the weights did not help at this data size; Sec. V-F).

Relevance R is the mean cross-encoder probability $\sigma(\ell_i)$ over E, where ℓ_i is the reranker logit for (q, e_i).

Coverage C is the fraction of the query's salient terms (lowercased alphanumeric tokens of at least three characters, excluding stop words) that occur in the concatenated evidence.

Agreement A is the mean over passage pairs (i, j) of the embedding cosine similarity s_{ij}, multiplied by 0.1 when the pair conflicts. Let N_i be the set of numbers in e_i. A pair conflicts if it is on the same topic ($s_{ij} > 0.55$) and $N_i \cap N_j = \emptyset$, or if both passages come from the same document, are near-duplicates ($s_{ij} > 0.9$), and $N_i \neq N_j$.

Trust/recency T is the mean over E of $0.7\, b_d + 0.3 \max\big(0, 1 - \min(a_d, 20)/20\big)$, where $b_d = 0.35$ if the source document d is marked obsolete (superseded by a newer version in the corpus) and 1 otherwise, and a_d is the document's age in years on 1 January 2024, the experiment's reference date. The constants in A and T were set by hand and are not tuned; only the thresholds (and, in the ablation, the weights) are fitted, inside training folds.

Conflict flag $X = 1$ if any same-document near-duplicate pair disagrees on a figure. A document cannot contradict itself, so $X = 1$ blocks a direct Answer: ECERAG drops both conflicting passages, re-scores the remainder, and answers only if it clears τ_A.

C. Selective Policy and Generation

ECERAG answers if $S \geq \tau_A$ and $X = 0$, asks for clarification if $\tau_C \leq S < \tau_A$, and abstains otherwise. In evaluation, Clarify counts as a decline (it lowers coverage and is never an error), since no user is present to answer the clarifying question. The generator is prompted to answer only from the evidence, cite passages by document tag, and reply `NOT_SUPPORTED` when the evidence lacks the answer (*grounded self-refusal*). For year-over-year financial questions, a deterministic calculator supplies the computed difference, since small generators made arithmetic errors on correctly retrieved figures. The corrective variant ECERAG+CR handles the Clarify band: the generator rewrites the query, the new top-5 is merged with E, conflicts are removed, the merged set is reranked against the original question, and the result is accepted only if its score clears τ_A and exceeds the original S by at least 0.05.

IV. Experimental Setup

A. Corpus and Benchmark

The corpus has six public documents (1,207 passages): Apple's Form 10-K for fiscal years 2022 and 2023 (a current/superseded pair), the NIST AI RMF 1.0, the OWASP Top 10:2021, and RFC 2616 and RFC 9110 (obsolete/current HTTP specifications). We hand-authored 60 questions verified against the sources: 15 single-hop, 15 multi-hop (most combining two document versions), 14 temporal/version-sensitive, and 16 unanswerable (false premise, out of corpus, or over-specific). Each answerable item has a gold answer and gold passage identifiers.

B. Perturbations

Each condition modifies the clean top-5 evidence of a question. *Irrelevant*: the lowest-ranked passage is replaced by the highest-scoring passage from a different document (55 applicable questions). *Contradictory*: the lowest-ranked slot is replaced by a copy of a gold passage with one number altered by ±37% (25 applicable: questions whose gold passage was retrieved and contains a number). *Counterfactual*: a passage from a current document is replaced by its nearest neighbor (embedding similarity) in the superseded version, i.e., what a stale index would return (46 applicable).

C. Systems

Unconditional RAG answers every question from E with the same prompt minus the refusal instruction. *Self-refusal RAG* uses the full prompt and declines only via `NOT_SUPPORTED`; this was the baseline of the pilot. *Evidence gate only* applies the S/X policy to the unconditional answers, isolating the sufficiency gate. *ECERAG* combines the gate with self-refusal and conflict handling; *ECERAG+CR* adds corrective retrieval. All systems share retrieval, reranking, evidence, and decoding, so they differ only in the decision step. Generators are Qwen2.5-Instruct models [22] with 0.5B, 1.5B, 3B, 7B, and 14B parameters, decoded greedily; the 14B model runs in 4-bit NF4 [23].

D. Calibration and Statistics

Thresholds are chosen inside training folds of a stratified 5-fold cross-validation repeated 20 times, so every question is scored out-of-fold. Candidate τ_A values are the observed training scores; we minimize risk + $0.5(1 - \text{coverage})$ subject to coverage ≥ 0.25, then choose τ_C on the ECERAG+CR outcome. Differences are reported with 95% paired bootstrap intervals (2,000 resamples of questions [24]), averaging over the 20 repeats within each resample.

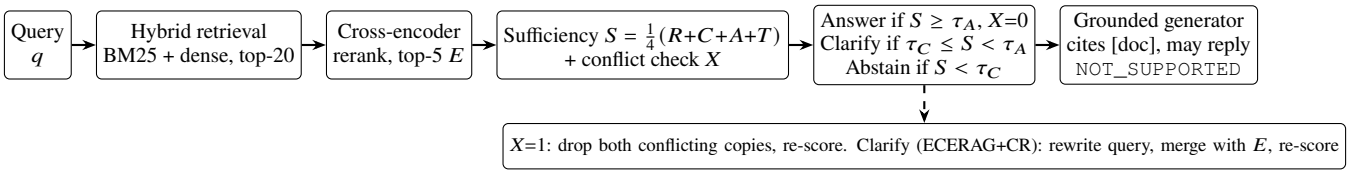


Fig. 1. ECERAG pipeline. Every component is specified in Sec. III.

E. Metrics and Grading

Coverage is the fraction of questions answered; *selective risk* is the error rate among answered questions, where any answer to an unanswerable question is an error; *abstention recall* is the fraction of unanswerable questions declined. The *risk–coverage AUC* [11] summarizes how well a score ranks answers (lower is better; a random ranking scores the overall error rate). Correctness is graded twice: (i) *lexically*, by recall of gold figures (with unit normalization, so that "$383.3 billion" matches "$383,285 million") and keywords; and (ii) by an *LLM judge*, Phi-3.5-mini-instruct [25], chosen from a different model family than the generators to avoid self-preference [17]. The graders agree on 75–82% of answers; the judge more often credits correct answers phrased differently from the gold text. We report the judge as primary, a choice the human check in Sec. V-G supports, and state where the lexical grader differs. The pilot's chunk-identifier context-relevance metric is not used: it scored retrieval at 0.09–0.10 even when the evidence held the answer, because the gold passage identifiers miss equivalent passages elsewhere. We instead check whether the gold answer appears in the retrieved evidence.

V. Results

A. Generator Scale

The generator is the dominant source of error at small scale. Self-refusal RAG's selective risk falls from 0.63 at 0.5B to 0.11 at 14B (judge; 0.59 to 0.26 lexical) on clean evidence (Fig. 2a), although the retrieved evidence contains the answer for 80% of questions at every scale. From 3B upward the generator declines all 16 unanswerable questions by itself; it also declines answerable ones (13 of 44 at 7B, 6 of 44 at 14B), over half of them multi-hop questions.

B. Decomposing the Gain

Table I compares the four decision rules at 7B and 14B, and Table II gives the paired differences from the unconditional baseline. ECERAG reduces selective risk substantially and significantly on clean, irrelevant, and outdated evidence, for both generators and both graders—for example from 0.43 to 0.13 at 7B and from 0.40 to 0.11 at 14B on clean evidence.

The decomposition shows where this gain comes from. Self-refusal alone reaches almost the same operating point: 0.13 at 7B and 0.11 at 14B; at 14B, ECERAG and self-refusal make identical decisions. Compared directly (Table II, right), ECERAG is not significantly different from self-refusal in any condition under the judge. Under the lexical grader it is better at 7B on outdated evidence ($[-0.28, -0.06]$) and slightly worse

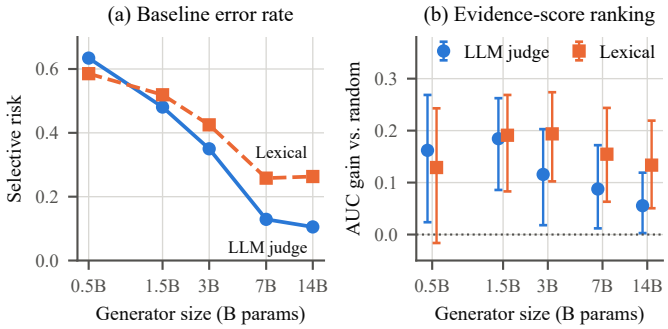


Fig. 2. (a) Selective risk of self-refusal RAG on clean evidence. (b) Gain of the sufficiency score's risk–coverage AUC over a random ranking, on the questions the generator answered; bars are 95% bootstrap intervals.

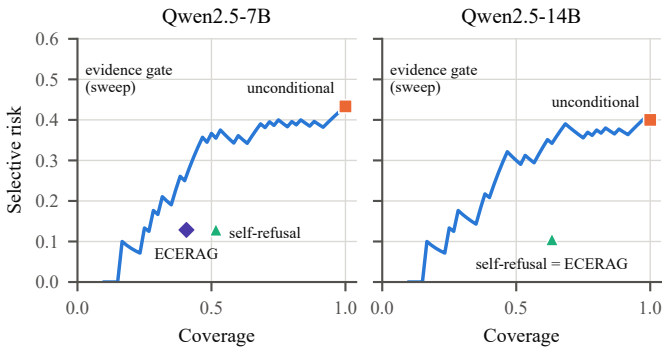


Fig. 3. Risk–coverage on clean evidence (LLM judge). The curve sweeps the evidence-gate threshold over unconditional answers; markers are cross-validated operating points.

at 14B on clean evidence ($[+0.00, +0.07]$). At 7B, matching self-refusal's risk costs ECERAG about 11 points of coverage. The evidence gate alone moves the unconditional system only slightly. Its reductions are significant under the judge for the 7B model on clean and outdated evidence and for the 14B model on outdated evidence (Table II); none is significant under the lexical grader. The gate also catches few unanswerable questions (abstention recall 0.09–0.33 versus 1.00 for self-refusal), because false-premise questions retrieve evidence that looks relevant. Fig. 3 shows the full trade-off: sweeping the gate threshold over unconditional answers traces a curve that lies well above the self-refusal operating point at every coverage.

C. A Worked Example

Table III follows one version-sensitive question through the 7B systems. On clean evidence every system that answers is

TABLE I

SELECTIVE RISK / COVERAGE (%) BY DECISION RULE, CROSS-VALIDATED OVER 60 QUESTIONS. UPPER BLOCK: LLM JUDGE; LOWER BLOCK: LEXICAL GRADER. LOWER RISK IS BETTER.

		Qwen2.5-7B			Qwen2.5-14B		
Grader	System	Clean	Irrelevant	Outdated	Clean	Irrelevant	Outdated
Judge	Unconditional RAG	0.43 / 100	0.47 / 100	0.63 / 100	0.40 / 100	0.38 / 100	0.48 / 100
	Evidence gate only	0.38 / 83	0.45 / 86	0.60 / 88	0.39 / 91	0.38 / 92	0.46 / 92
	Self-refusal RAG	0.13 / 52	0.16 / 58	0.36 / 54	0.11 / 63	0.03 / 62	0.24 / 54
	ECERAG	0.13 / 41	0.14 / 41	0.34 / 43	0.11 / 63	0.03 / 62	0.24 / 54
Lexical	Unconditional RAG	0.52 / 100	0.51 / 100	0.57 / 100	0.47 / 100	0.40 / 100	0.46 / 100
	Evidence gate only	0.51 / 74	0.50 / 74	0.54 / 77	0.46 / 91	0.40 / 92	0.46 / 93
	Self-refusal RAG	0.26 / 52	0.25 / 58	0.40 / 54	0.26 / 63	0.21 / 62	0.24 / 54
	ECERAG	0.23 / 34	0.20 / 35	0.24 / 35	0.30 / 53	0.23 / 50	0.26 / 47

TABLE II

CHANGE IN SELECTIVE RISK: 95% PAIRED BOOTSTRAP INTERVALS (2,000 RESAMPLES OF QUESTIONS). LEFT AND MIDDLE: RELATIVE TO UNCONDITIONAL RAG. RIGHT: ECERAG RELATIVE TO SELF-REFUSAL RAG, I.E., WHAT THE EVIDENCE GATE ADDS ON TOP OF THE GENERATOR'S OWN REFUSAL. INTERVALS EXCLUDING ZERO ARE IN BOLD. †IDENTICAL DECISIONS: WITH JUDGE-CALIBRATED THRESHOLDS, THE 14B GATE NEVER DECLINES A QUESTION THAT SELF-REFUSAL ANSWERS.

		Gate only − Unconditional		ECERAG − Unconditional		ECERAG − Self-refusal	
Model	Evidence	Judge	Lexical	Judge	Lexical	Judge	Lexical
7B	Clean	**[−.11,−.01]**	[−.05,.04]	**[−.45,−.16]**	**[−.45,−.14]**	[−.03,.04]	[−.13,.06]
	Irrelevant	[−.07,.01]	[−.04,.03]	**[−.47,−.21]**	**[−.45,−.16]**	[−.09,.04]	[−.15,.05]
	Outdated	**[−.07,−.01]**	[−.06,.02]	**[−.47,−.11]**	**[−.50,−.16]**	[−.10,.06]	**[−.28,−.06]**
14B	Clean	[−.04,.02]	[−.04,.02]	**[−.42,−.17]**	**[−.32,−.02]**	[.00,.00]†	**[.00,.07]**
	Irrelevant	[−.02,.02]	[−.02,.03]	**[−.49,−.22]**	**[−.32,−.02]**	[.00,.00]†	[−.01,.06]
	Outdated	**[−.05,−.00]**	[−.02,.03]	**[−.40,−.09]**	**[−.36,−.06]**	[.00,.00]†	[−.01,.06]

correct. Under outdated evidence, the stale index replaces the only RFC 9110 passage with its nearest RFC 2616 neighbor, so all five passages come from the obsolete specification, and the unconditional system inverts the answer, naming the obsolete specification as authoritative. Self-refusal declines instead. The sufficiency score registers the stale source (T falls from 0.39 to 0.24), but a rise in agreement leaves S almost unchanged, and S sits below the typical cross-validated τ_A (0.67) in both conditions. ECERAG therefore asks for clarification on both, which avoids the error but also withholds the correct clean answer. The example shows the pattern of Sec. V-B in miniature: grounded refusal catches the stale case, while the gate trades coverage for a protection that self-refusal already provides.

D. The Sufficiency Score as a Ranking Signal

Although thresholding S adds little to self-refusal, S ranks answers well. These AUCs involve no calibration data: S uses fixed equal weights and the single signals have no parameters, so nothing is fitted and every question is effectively a test item (the fitted variants in Sec. V-F are scored out-of-fold). Restricted to the questions the generator answered (so no credit comes from its own refusals), the risk–coverage AUC of S beats a random ranking at every scale from 1.5B to 14B under both graders, with bootstrap intervals that exclude zero, narrowly so for the 14B judge (lower bound 0.003; Fig. 2b). At 7B, S reaches an AUC of 0.04 against 0.13 for a random ranking (judge; 0.10 against 0.26 lexical). Table IV breaks the score down: relevance and coverage carry most of the signal, trust/recency helps, and agreement is weakest on its own.

TABLE III

WORKED EXAMPLE (QWEN2.5-7B, Q055): "DOES RFC 9110 OR RFC 2616 CURRENTLY TAKE PRECEDENCE FOR DEFINING HTTP SEMANTICS, AND WHICH ONE SHOULD BE TREATED AS HISTORICAL ONLY?" GOLD: RFC 9110; RFC 2616 IS HISTORICAL.

	Clean evidence	Outdated evidence
Signals	R=.42, C=.83, A=.36, T=.39; S=.50	R=.42, C=.75, A=.53, T=.24; S=.49
Unconditional	RFC 9110 takes precedence; RFC 2616 is historical. **Correct**	"RFC 2616 takes precedence… RFC 9110 should be treated as historical." **Wrong**
Self-refusal	Same as unconditional. **Correct**	`NOT_SUPPORTED`. **Declined**
ECERAG	$S < \tau_A$: Clarify. **Declined**	$S < \tau_A$: Clarify. **Declined**

E. Contradictory Evidence and Corrective Retrieval

The conflict check flagged all 25 contradictory evidence sets and 1 of 60 clean ones (the pilot's rule could not detect a copy with one figure changed). Acting on the flag is harder. Refusing every flagged case yields zero risk at zero coverage. Dropping both conflicting copies recovers 38% (7B) and 28% (14B) coverage, but at a selective risk of 0.27 and 0.14 (judge), not significantly different from the unconditional baseline's 0.20 and 0.12, because the dropped pair usually contains the passage with the answer. Self-refusal RAG remains strong on this condition (0.14 and 0.05 risk at 88% coverage). Corrective retrieval that replaces the evidence wholesale increased risk

TABLE IV
RISK–COVERAGE AUC OF EACH SIGNAL ON ANSWERED QUESTIONS (JUDGE / LEXICAL). LOWER IS BETTER; "RANDOM" IS THE EXPECTED AUC OF A RANDOM RANKING.

Score	1.5B	7B	14B
Random	.48 / .52	.13 / .26	.11 / .26
S (equal weights)	.30 / .33	.04 / .10	.05 / .13
R relevance	.28 / .33	.04 / .12	.05 / .15
C coverage	.26 / .29	.05 / .09	.05 / .14
A agreement	.45 / .53	.09 / .25	.05 / .25
T trust/recency	.39 / .39	.05 / .11	.05 / .16

at 14B (0.05 to 0.23 on contradictory evidence); the merged, margin-gated version left coverage and selective risk unchanged in every condition.

F. Ablations

Fitting the four weights by logistic regression, and adding an NLI-based grounding score of the drafted answer, did not significantly improve selective risk over equal weights at 7B (judge, 95% interval of the difference from the baseline $[-0.10, +0.03]$). The grounding score's apparent AUC advantage disappeared once the generator's own refusals were excluded from the ranking. Removing any single signal changed clean-condition risk by at most 0.03. With the pilot's fixed-split calibration (17 calibration and 43 test questions), the grid selected $\tau_A = 0.80$ for both 7B and 14B, giving zero errors at 26% coverage against 0.13 and 0.10 for the baseline: a conservative operating point that the cross-validated cost function does not choose.

G. Human Validation of the Graders

An annotator labeled 100 answers from the 14B generator, sampled across question types and all four evidence conditions (28 clean, 27 irrelevant, 23 outdated, 22 contradictory). The annotator was blind to the automatic grades and followed a written rubric that judges correctness against the gold answer, faithfulness against the evidence shown, and citations against the cited documents. In a first pass, the LLM judge agreed with the human on 91% of answers (Cohen's $\kappa = 0.52$; 5 lenient and 4 strict disagreements); a rubric-guided recheck of these 9 disagreements resolved all of them in the judge's favor. The lexical grader agreed on 80% ($\kappa = 0.31$), mostly by rejecting 16 answers the human accepted, which confirms that it under-credits correct answers phrased differently from the gold text (κ is depressed by the 89% base rate of correct answers). The human found 98% of answers faithful to the evidence but only 74% correctly cited, mostly because the answer carried no citation at all (23 of the 26 failures). The reductions relative to unconditional RAG (about 0.30) are therefore not an artifact of grading. The smaller gate-only effects (at most 0.11) remain the least certain result, because of sample size rather than grader error.

VI. DISCUSSION

Implications. For enterprise RAG, the most effective single control in our study is to instruct the generator to answer only from cited evidence and to refuse otherwise, provided the generator is large enough to follow the instruction (3B and up here). Evidence-sufficiency scores add a calibrated, inspectable confidence signal on top of that. They are useful for ranking, triage, and routing outdated or conflicting evidence to review, and they provide an audit trail that self-refusal alone does not. Whether to gate on them is a risk-tolerance choice, which Fig. 3 makes explicit.

Limitations. The benchmark has 60 questions from six documents, so many intervals are wide; results may not transfer to other domains without recalibration. Correctness is graded automatically. The human check (Sec. V-G) used a single annotator, so inter-annotator agreement is not measured, and it covers only the 14B model. Because that model declined all 16 unanswerable questions, the sample contains none of them, so the grading of answers to unanswerable questions is not human-checked. Unanswerable questions were authored by us and may be easier to detect through self-refusal than those arising in practice. All generators come from one model family, and the 14B model was quantized. We do not compare the sufficiency score with generator-side confidence (token probabilities or semantic entropy [13]) as a ranking signal. The perturbations are synthetic, and the conflict check matches our injection procedure, which is only one kind of real-world conflict.

VII. CONCLUSION

Enterprise RAG should move from retrieve-then-generate to retrieve–assess–decide. In a controlled study across five generator sizes, ECERAG reduced the error rate on answered questions by roughly two-thirds relative to unconditional RAG, a result that a human check of the grading supports. Almost all of the reduction comes from grounded self-refusal by the generator; the evidence-sufficiency gate adds a small, occasionally significant improvement, and its score reliably ranks answer correctness from 1.5B to 14B parameters. Future work will add a second annotator, another model family and more realistic unanswerable questions, and will use the sufficiency score to drive targeted retrieval rather than blanket abstention.

CODE AND DATA AVAILABILITY

The pipeline, corpus sources, benchmark, raw model outputs, and analysis scripts are available at https://github.com/yomisys/LLM.

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